

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2			Method Diagram of charging circuit. Description of how to charge. Diagram of discharging circuit. Description of how to discharge. Collection of data described i.e. every 20 s measure V etc Valid meter correctly placed. Analysis A discharging equation given e.g. $I = I_0 e^{-\frac{t}{RC}}$ or $\ln V = \ln V_0 - \frac{t}{RC}$ Correct graph stated e.g. $I \propto t$, $V \propto t$, $Q \propto t$, $\ln V \propto t$. Valid method for time constant e.g. 37% left OR $\frac{-1}{\text{gradient}}$ Shape of graph described / shown correctly (i.e. exponential decay shape OR straight and negative gradient). Statement that time constant = RC (for comparison etc)	6			6	1	6

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			5-6 marks Comprehensive description of both the method and analysis. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Comprehensive description of either the method or analysis or limited description of both provided. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks Limited description of either the method or analysis <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
			Question 2 total	6	0	0	6	1	6